



Jianpi Qushi Heluo formula ameliorates podocytes injury related with ROS-mediated NLRP3 inflammasome activation in membranous nephropathy by promoting PINK1-dependent mitophagy

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ABSTRACT

Ethnopharmacological relevance: Jianpi Qushi Heluo formula (JQHF) is an evidence-based herbal formula based on "Fangji Huangqi Decoction" in the classic of traditional Chinese medicine (TCM) synopsis of the Golden Chamber. Its effectiveness in reducing idiopathic membranous nephropathy (IMN) proteinuria, edema and other clinical symptoms and improving kidney injury has been confirmed by clinical trials and animal experiments. These results demonstrate the critical significance of reducing podocyte injury by regulating mitophagy through integrating ethnopharmacology.

Aim of the study: This study aimed to explore the protective effects of JQHF on podocyte injury in IMN, focusing on its role in promoting PINK1-dependent mitophagy to inhibit reactive oxygen species (ROS) mediated activation of the NOD-like receptor family pyrin domain-containing 3 (NLRP3) inflammasome.

Materials and methods: The main components of JQHF were identified by ultra-performance liquid chromatography combined with quadrupole time-of-flight mass spectrometry (UPLC-Q-TOF/MSE). Passive Heymann nephritis (PHN) was induced in rats using anti-Fx1A antiserum, while podocyte injury in vitro was stimulated with sublytic complement C5b-9 (sC5b-9). The PHN rats were treated with JQHF, and autophagy inhibitor 3-methyladenine (3-MA) combined with JQHF was used for pathway verification, and Benazepril served as a positive control. In vitro, podocytes were exposed to palmitic acid (PA, a ROS inducer) and N-acetylcysteine (NAC, a ROS scavenger). Further, we silenced the expression of PINK1 in podocytes model and intervened with JQHF-containing serum. Renal function was assessed through biochemical analyses and histopathology. Mitochondrial function was measured by detecting mitochondrial membrane potential (MMP), ROS levels as well as mitochondrial ultrastructure. The expression of podocyte structural proteins (desmin, nephrin, podocin) and inflammasome-related markers (NLRP3, caspase-1, IL-1 β , IL-6) was analyzed to assess podocyte injury and inflammasome activation.

Results: (1) In vitro, in contrast to the control group, PA intervention caused increased ROS accumulation, reduced MMP, upregulated NLRP3 and caspase-1 expression, as well as elevated expression of inflammatory factors IL-1 β and IL-6. The expression of nephrin and podocin was notably reduced. By contrast, NAC reversed

Abbreviations: ALB, albumin; BSA, bovine serum albumin; CS, control serum; DAB, diaminobenzidine (DAB); DCFH-DA, dichlorofluorescein diacetate; ESRD, end-stage renal disease; JQHF, Jianpi Qushi Heluo Formula; JQHFS, JQHF-containing serum; HE, hematoxylin-eosin; IMN, idiopathic membranous nephropathy; IFN- γ , γ -Interferon; IOD, integral optical density; MMP, mitochondrial membrane potential; mtROS, mitochondrial ROS; mtDNA, mitochondrial DNA; NAC, N-acetylcysteine; NLRP3, NOD-like receptor family pyrin domain-containing 3; PA, palmitic acid; PAS, periodic acid-Schiff; PHN, passive heyman nephritis; ROS, reactive oxygen species; sC5b-9, sublytic complement C5b-9; TC, total cholesterol; TCM, traditional Chinese medicine; TEM, transmission electron microscopy; UPLC-Q-TOF/MSE, ultra-performance liquid chromatography quadrupole-time of flight mass spectrometry in elevated energy mode; 24h UTP, 24-h urinary protein; 3-MA, autophagy inhibitor 3-methyladenine.

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these effects. (2) In vivo, JQHF effectively ameliorated mitochondrial damage, reduced NLRP3 expression and mitigated podocyte injury in PHN rats. The protective effects of JQHF were diminished by 3-MA, confirming the involvement of autophagy. (3) In vitro, JQHF-containing serum attenuated C5b-9-induced podocyte injury, improved mitochondrial dysfunction, and inhibited ROS-mediated activation of NLRP3 inflammasome. Silencing PINK1 significantly reversed these protective effects.

Conclusion: JQHF can alleviate podocyte damage, that through inhibiting ROS-mediated NLRP3 inflammasome activation by improving PINK1-mediated mitophagy.

1. Introduction

Idiopathic membranous nephropathy (IMN), one of the most prevalent forms of primary glomerular diseases, has increased at an annual rate of 13 % over the past 11 years (Xu et al., 2016). Approximately one-third of untreated patients round into end-stage renal disease (ESRD) within 10–15 years of onset, posing a significant public health concern (Hou et al., 2018). Current standard treatments primarily involve immunosuppressive and cytotoxic drugs, which are frequently correlated with adverse effects such as poor tolerance, high recurrence rates, and nephrotoxicity (Bi et al., 2024; Scolari et al., 2021). Therefore, exploring new therapeutic targets and finding safer and more effective treatment plans are still urgent issues.

IMN is an antigen-antibody reaction caused by the combination of circulating autoantibodies with target antigens on the surface of glomerular podocytes. Podocytes are critical components of the glomerular filtration barrier, and their injury exerts a pivotal role in IMN pathogenesis (Avasare et al., 2024; Zhang et al., 2023). Immune complex deposition beneath podocytes triggers activation of complement, leading to the formation of the membrane attack complex (C5b-9), which integrates into podocyte membranes and promotes the generation and accumulation of reactive oxygen species (ROS). Numerous studies have elucidated the detrimental effect of excessive ROS on podocyte damage (Qin et al., 2020; Wang et al., 2022). The prior research of our team confirmed that ROS accumulation in podocytes induces mitochondrial dysfunction, a key contributor to IMN progression (Wang et al., 2021). Mitochondria are the power supply of cells and participate in important cellular functions, including ATP production, intracellular Ca²⁺ adjustment, ROS generation and clearance, and activation of caspase family proteases and so on (Chen et al., 2023; Dong et al., 2025). The preliminary research of our team found mitochondrial dysfunction leads to podocyte apoptosis in IMN rats.

Moreover, mitochondrial health is closely tied to the activation of the NOD-like receptor family pyrin domain-containing 3 (NLRP3) inflammasome, particularly in rheumatic, metabolic, and neurodegenerative diseases (Vande and Lamkanfi, 2024). The NLRP3 inflammasome, comprising the protein NLRP3, ASC, and pro-caspase-1, is recognized a central link that initiates local inflammatory response (Wang et al., 2023). Mitochondrial dysfunction can trigger NLRP3 activation through mechanisms such as excessive mitochondrial ROS (mtROS) production, cytosolic translocation of mitochondrial DNA (mtDNA), and mitochondrial relocation near NLRP3 via α -tubulin acetylation (Boini et al., 2016; Gurung et al., 2015; Misawa et al., 2013; Nakahira et al., 2011). Elevated NLRP3 expression was observed in renal biopsy tissues in IMN patients, which correlated with disease progression (Vilaysane et al., 2010). Additionally, studies have observed overexpression of NLRP3, IL-1 β , and caspase-1 in IMN podocytes, associating these changes with podocyte injury. However, it is unclear whether the activation of NLRP3 correlated with mitochondrial dysfunction mediated by ROS accumulation in podocytes (Liu et al., 2019). This research aims to expound the relationship between mitochondrial dysfunction and NLRP3 inflammasome activation in IMN-associated podocyte injury and to explore potential therapeutic strategies.

Over the past decade, there has been a remarkable leap in clinical understanding of the structure and function of mitochondria. In the field of chronic kidney disease (CKD), research focused on the regulation of

mitophagy has also achieved breakthrough progress. As a key intervention method, regulating mitophagy can not only significantly improve mitochondrial dysfunction but also effectively alleviate oxidative stress damage to cells, making it an important strategy with great application prospects in the treatment of CKD. Mitophagy, an optional form of autophagy that removes impaired and dysfunctional mitochondria, holds a key protective mechanism for maintaining mitochondrial homeostasis and reducing ROS accumulation. PINK1/Parkin is currently extensively studied pathways mediating mitophagy. (Han et al., 2023; Nguyen et al., 2016). Enhancing PINK1/Parkin-mediated mitophagy exerts positive effects on CKD. For example, PINK1 or Parkin knockout mice exhibit exacerbated renal dysfunction, tissue damage, and apoptosis following cisplatin exposure. Similarly, in palmitic acid (PA)-induced podocyte injury models, mitochondrial dysfunction and mtROS accumulation are aggravated by Parkin silencing, leading to increased podocyte apoptosis (Jiang et al., 2020; Li et al., 2017; Wang et al., 2018). These findings suggest that PINK1-mediated mitophagy plays a critical role in podocyte protection and represents a promising therapeutic target for IMN.

Jianpi Qushi Heluo formula (JQHF), an evidence-based formula, is one of the most commonly used TCM formulations for IMN. The formulation principle derives from "Fangji Huangqi Decoction", a classic prescription from synopsis of the Golden Chamber, combining lidongyuan's "Yin Fire" theory. In traditional clinical practice, this formula exhibits the therapeutic effects of fortifying the spleen and tonifying the kidney, resolving dampness and dispelling blood stasis, and is clinically applied to ameliorate renal edema and foamy urine (Wang et al., 2022; Zeng et al., 2023). JQHF is composed of 9 herbs (Table 1), contemporary medical studies have revealed the pharmacological mechanism of these herbs to alleviate kidney injury. For instance, *Astragalus membranaceus* (Fisch.) Bge. can significantly decrease the urinary protein of diabetic nephropathy mice and improve podocyte apoptosis, in vitro research results show that its podocyte protective mechanism is to enhance autophagy via suppressing pi3k/akt/mtor signal transmission pathway (Yang et al., 2020). The combination of *Angelica sinensis* (Oliv.) Diels and *Astragalus membranaceus* (Fisch.) Bge. can alleviate renal interstitial fibrosis, and reduce renal oxidative stress by regulating AGEs/RAGE pathway in diabetic nephropathy mice (Zhang et al., 2024). The combination of *Atractylodes macrocephala* Koidz. and *Poria cocos* (Schw.) Wolf has been proven to improve renal function and injury. The mechanism involves inhibiting TLR4/myd88 pathway, NLRP3 inflammasome activation as well as regulating Wnt4/beta catenin signaling (La et al.,

Table 1
The composition of JQHF.

Chinese Name	Latin name of botanical plant	Part used	Dosage (g)
Huangqi	<i>Astragalus membranaceus</i> (Fisch.) Bge.	Root	30
Baizhu	<i>Atractylodes macrocephala</i> Koidz.	Rhizome	15
Fuling	<i>Poria cocos</i> (Schw.) Wolf	Rclerotium	30
Fangji	<i>Stephania tetrandra</i> S.Moore	Root	20
Zisuye	<i>Perilla frutescens</i> (L.) Britt.	Roliage	15
Chuanshanlong	<i>Dioscorea nipponica</i> Makino	Rhizome	30
Renshen	<i>Panax ginseng</i> C. A. Mey.	Rhizome	10
Danggui	<i>Angelica sinensis</i> (Oliv.) Diels	Root	15
Heye	<i>Nelumbo nucifera</i> Gaertn.	Roliage	15

2018; Yang et al., 2015). *Perilla frutescens* (L.) Britt. can improve the proteinuria of IgA mice and reduce the deposition of serum IgA, glomerular IgA and IgG (Makino et al., 2003). Ginseng leaves are rich in ginsenosides, studies have found that ginsenoside RB3 can alleviate cisplatin induced renal injury, and its mechanism may be through AMPK/mTOR-dependent autophagic regulation (Xing et al., 2019). The previous studies of our team demonstrated that JQHF achieved an 80 % clinical remission rate in IMN patients after six months of treatment, with minimal side effects, even in refractory cases (Shi et al., 2018). A single-center cohort study involving 140 IMN patients further confirmed that JQHF significantly reduced proteinuria levels (Lang et al., 2020). Mechanistic investigations demonstrated the accumulation of ROS and mitochondrial dysfunction in podocytes and rat models of IMN (Wang et al., 2021). Importantly, JQHF was found to upregulate the PINK1/-Parkin pathway, eliminating damaged mitochondria, and reducing ROS accumulation.

In this study, a PHN rat model was established using anti-Fx1A antiserum, and podocyte injury model was stimulated by complement C5b-9 in vitro. We focused on the interaction between mitochondrial dysfunction and ROS-mediated NLRP3 inflammasome activation in IMN-related podocyte injury. Furthermore, based on the above research exploring the podocyte protective mechanism of JQHF.

2. Materials and Methods

2.1. Preparation of JQHF

The specific herbs of JQHF are displayed in Table 1. All herbs were procured from Hebei Baicaokang Pharmaceutical Co., Ltd. (Anguo City, Hebei, China; Certificate No. JIY20190064). Based on body surface area conversion factors between humans and animals the corresponding daily dose for rats was calculated to be 16.2 g/kg. To evaluate the dose-dependent effects of JQHF, a higher dose of 32.4 g/kg was also included in this study. These nine herbs were processed into decoctions by Xiyuan Hospital, including two concentrations of 1.62 g/mL and 3.24 g/mL. These were administered orally to the rats via gavage.

2.2. UPLC-Q-TOF/MSE exploration of JQHF

The UPLC-Q-TOF/MSE method was employed to identify the primary chemical constituents of JQHF. A 0.5 g sample of JQHF was extracted with methanol at a 1:3 ratio (w/v). The sample mixture underwent ultrasonic processing for 10 min followed by centrifugation. The resulting supernatant was collected and diluted tenfold for analytical purposes. Chromatographic separation was achieved using an Acquity UPLC system interfaced with a Synapt G2-Si HDMS high-resolution mass spectrometer (Waters Corporation, USA).

2.3. Animals and treatment administration

All experimental procedures involving animals were conducted in strict adherence to the ARRIVE guidelines (Animal Research: Reporting of In Vivo Experiments). The study protocol received formal approval from Animal Ethics Committee of Xiyuan Hospital, China Academy of Chinese Medical Sciences (Beijing, China; approval numbers: 2019XLC020-2 and 2023XLC006-2).

Male Sprague-Dawley (SD) rats were procured from Beijing Vital River Laboratory Animal Technology Co., Ltd. (Beijing, China) and maintained under specific pathogen-free conditions with regulated ambient temperature (25 ± 1 °C), a 12/12-h light-dark cycle, and 50 ± 5 % moisture level. Following a 3-day acclimatization period, thirty-two rats were intravenously administered sheep anti-rat Fx1A serum (PTX-002S, Probetex, San Antonio, USA) via the tail vein to induce PHN model. The injection dose was 0.5 mL/100 g. An additional six rats received an equivalent volume of sterile physiological saline and served as the healthy control cohort. Two weeks post-injection, model

establishment was confirmed through quantitative analysis of 24-h urinary protein excretion (24 h UTP) using an automated biochemistry analyzer (Ccbas 8000, Roche Diagnostics GmbH, Mannheim, Germany). Rats demonstrating significant proteinuria (24 h UTP >10 mg) were stratified into four experimental cohorts (n = 6 per group): the model group was administered an equivalent volume of distilled water, while the benazepril group was administered benazepril (Nuohua, Beijing, China) at a dose of 1 mg/100 g. The JQHF groups were intervened with JQHF at doses of 1.62 g/100 g (JQHF-M) or 3.24 g/100 g (JQHF-H). The dose in the JQHF-M group corresponded to the clinical dosage used in patients. The normal group was administered an equivalent volume of distilled water. Treatments were administered via oral gavage daily for 6 weeks. At the 6-week endpoint, whole blood was obtained and centrifuged for biochemical assays. Following euthanasia, kidney tissues were harvested for pathological examination.

2.4. Biochemical index detection

24 h UTP was performed every 2 weeks throughout the study. Before sacrifice, all rats were kept fasting for 12 h. Following anesthesia with isoflurane, blood specimens were obtained from an abdominal segment of the aorta, subsequently were centrifuged and analyzed for hepatic function markers, renal function parameters, as well as lipid metabolism profiles using an automated clinical chemistry analyzer (Ccbas 8000, Roche Diagnostics, Germany).

2.5. Cell culture and induction of C5b-9 membrane attack complex formation in vitro

Mouse podocyte cell line MPC-5 (FH1009, Fu Heng BioLogY, Shanghai, China) was cultured of 10 U/mL mouse recombinant interferon- γ (IFN- γ , Pepro Tech, England) at 33 °C to promote proliferation. When the cells reached 80–90 % confluence, they were incubated at 37 °C without IFN- γ for 14 days to induce differentiation.

The sC5b-9 attack model was implemented based on formerly validated approach (Chen et al., 2010): Differentiated podocytes were preincubated with C5b-6 (CPT-A122, CompTech, Texas, USA, 0.8 μ g/mL) and C7 (CPT-A124, CompTech, Texas, USA, 10 μ g/mL) for 15 min at 37 °C to initiate C5b-9 assembly. Subsequently, C8 (CPT-A125, CompTech, Texas, USA, 10 μ g/mL) and C9 (CPT-A126, CompTech, Texas, USA, 10 μ g/mL) were introduced, and incubation continued for another 30 min. C5b-9 deposition on podocyte membranes was confirmed using confocal microscopy (LSM 510, Carl Zeiss, Jena, Germany) and flow cytometry (FACS Aria, BD, San Jose, CA, USA) with an anti-C5b-9 antibody (SC-66190, Santa Cruz Biotechnology, USA).

2.6. Preparation of drugs and experimental groups in vitro

JQHF-containing serum (JQHFS) was prepared as follows: Fifteen SD rats were orally administered JQHF at a concentration of 1.62 g/100 g for 7 days, another fifteen rats received normal saline (1 mL/100 g) as the control serum (CS). After 7 days, blood was collected from each rat and centrifugation was performed (3000 rpm, 10 min, 4 °C). The resulting serum was filtered, subjected to heat inactivation (56 °C, 30 min), and cryopreserved at -20 °C for subsequent use. Palmitic acid (PA) (161420, Aladdin, Shanghai, China) and N-acetylcysteine (NAC) (S0077, Beyotime, Shanghai, China) were prepared following earlier published protocol (Qin et al., 2019; Wang et al., 2022).

The differentiated podocytes were systematically allocated into the following groups: (1) Control group with 10 % CS, (2) Model group with 10 % CS, (3) JQHF group with 10 % JQHFS, (4) PA group with 100 μ mol/L palmitate, (5) NAC group with 10 μ mol/L NAC, (6) shRNA-PINK1 group, (7) JQHF + shRNA-PINK1 group. In addition to the control group, c5b-6, C7, C8 and C9 were incubated in other groups.

2.7. shRNA-PINK1 transfection

MPC5 cells were cultured, and the shRNA-PINK1 gene was cloned into the pENTR™/MHI/TO vector to construct the pENTR™/MHI/TO-shRNA-PINK1 recombinant plasmid. Transfection of the shRNA-PINK1 plasmid into podocytes was performed using Liposome 2000 (HY-112764, MCE, Shanghai, China). The procedure was as follows: One day prior to transfection, podocytes were seeded into six-well plates. Two clean, sterile centrifuge tubes were prepared. Into each tube, 125 µL of antibiotic and serum-free DMEM culture medium was added. To the first tube, 100 pmol of siRNA was added, and to the second tube, 5 µL of Lipo6000™ Transfection Reagent (C0526, Beyotime, Shanghai, China) was added. The mixtures were carefully agitated and subsequently incubated at ambient temperature for 5 min. Next, the siRNA-containing solution was gently added to the tube containing the Lipo6000™ Transfection Reagent solution. The mixture was gently inverted or pipetted to ensure thorough mixing and left to stand for another 5 min at room temperature (stable within 6 h of storage at room temperature). Subsequently, 250 µL of the transfection complex was dispensed into a 6-well culture plate and incubated for 48 h. The cells were collected before and after transfection for analysis by WB and PCR. The knock-down efficiency of PINK1 was verified to be 80 %, which can be used for subsequent experiments.

2.8. Assessment of ROS generation in vitro

ROS levels were detected by mitochondrial ROS detection kit (S0033S, Beyotime, Shanghai, China) and the ROS-sensitive fluorescent probe 2',7'-dichlorodihydro-fluorescein diacetate (DCFH-DA). Podocytes were incubated with 10 µM DCFH-DA at 37 °C for 20 min in the dark. After thorough washing, PBS was added, and the fluorescence intensity of each group was detected under the fluorescence enzyme marker.

2.9. Flow cytometry analysis

Rats were euthanized, and fresh kidney tissue was immediately dissected. A homogeneous cell suspension was prepared from rat kidney cortical tissue using gentle physical dissociation and 40 µm mesh filtration. The cell density was normalized to 1×10^6 cells/mL in the final suspension. For the analysis of intracellular ROS, MMP, and cell apoptosis, the cells were incubated with DCFH-DA (S0033, Beyotime, Shanghai, China) and tetrachloro-tetraethylbenzimidazol carbocyanine iodide (JC-1) (C2006, Beyotime, Shanghai, China) for 30 min. After incubation, harvested cells were subjected to analysis using a flow cytometer (EPICS ELITE, Beckman Coulter, Inc., Los Angeles, USA).

2.10. Histopathological assessment of kidney specimens

The renal histopathological examination was conducted in accordance with previously published methods of our team (Wang et al., 2022): in short, the renal specimens were immediately fixed, subsequently dehydrated, embedded, sectioned, and stained with hematoxylin-eosin (HE) and periodic acid-Schiff (PAS) for comprehensive morphological evaluation. The renal pathological evaluation encompassed assessments of tubular atrophy, dilation, tubular morphology, interstitial inflammation, and fibrosis. Six randomly selected fields per renal section were examined at $\times 100$ magnification. A standardized semiquantitative grading system (0–3 scale) was applied to quantify tubular injury severity. Analysis was implemented by Image-Pro Plus 6.0 software (Media Cybernetics, Inc., Rockville, USA).

2.11. Transmission electron microscopy (TEM) examination

Renal tissue samples approximately 1 mm^3 were preserved in 2.5 % glutaraldehyde solution for fixation, followed by dehydration,

infiltration with acetone, embedding in epoxy resin, and sectioning into ultrathin slices. The sections were stained with uranyl acetate and lead citrate. The samples were then examined under a transmission electron microscope (JEM-1230, JEOL Japan Electronics Co., Ltd., Tokyo, Japan). Special attention was given to subepithelial electron-dense deposits and podocyte mitochondrial alterations.

2.12. Immunohistochemistry staining

The kidney tissue sections underwent deparaffinization and rehydration, followed by antigen retrieval in 10 mM citrate buffer (pH 6.0). After blocking with 3 % BSA, samples were incubated with anti-Nephrin primary antibodies (ab216341, Abcam, Cambridge, UK). Subsequent PBS washes preceded incubation with biotinylated secondary antibodies at room temperature. Staining was performed using 3,3'-diaminobenzidine (DAB) and hematoxylin. Nephrin expression was quantified by measuring the integral optical density (IOD) per field using Image-Pro Plus 6.0 software.

2.13. Western blot analysis

Renal cortex tissues and podocytes were homogenized using a commercial protein extraction kit (P0033, Beyotime, Shanghai, China). Mitochondrial and cytoplasmic fractions were isolated with a specialized kit (C3606, Beyotime, Shanghai, China). Protein concentrations were determined via BCA assay (P1511, Applygen, Beijing, China). Following electrophoretic separation, protein bands were electroblotted onto nitrocellulose substrates. After a 1 h blocking with 5 % BSA, membranes were probed overnight at 4 °C with the primary antibodies as follows: anti-NLRP3 (ab283819, Cambridge, UK), anti-Caspase-1 (ab138483, Abcam, Cambridge, UK), anti-IL-1 β (ab315084, Abcam, Cambridge, UK), and anti-IL-6 (ab290735, Abcam, Cambridge, UK). After washing, membranes were incubated with HRP-conjugated secondary antibodies. Protein-antibody interactions were visualized using an ECL detection system (P1010, Applygen, Beijing, China) and captured with a multifunctional imager. Image-Pro Plus 6.0 software was applied to quantify the protein bands.

2.14. Immunofluorescence staining

The immunofluorescence staining was conducted derived from previously published methods of our team (Wang et al., 2022): firstly, the paraffin-processed renal cortical tissues were sectioned into 3-µm-thick slices. Next, tissue sections were saturated with 3 % BSA and sequentially incubated with either FITC-conjugated anti-rat IgG (4416s, CST, Boston, USA) or primary anti-C5b-9 antibody (bs-2673R, Bioss, Beijing, China), followed by FITC-labeled goat anti-rabbit IgG secondary antibody (ab6717, Abcam, Cambridge, UK). Fluorescence images were acquired using an Olympus TH4-00 inverted microscope. Glomerular deposition of IgG and C5b-9 immune complexes was quantified through Image-Pro Plus 6.0 software analysis.

The podocytes were fixed in chilled 4 % paraformaldehyde (4 °C, 20 min), followed by PBS washes and membrane permeabilization for 20 min. After antigen retrieval and blocking with 3 % BSA for 30 min, samples were incubated overnight at 4 °C with primary antibodies: anti-NLRP3 (ab283819, Cambridge, UK), anti-Caspase-1 (ab138483, Abcam, Cambridge, UK), anti-IL-1 β (ab315084, Abcam, Cambridge, UK), anti-IL-6 (ab290735, Abcam, Cambridge, UK), anti-Podocin (20384-1-AP, Proteintech, Wuhan, China), and anti-Desmin (ab227651, Abcam, Cambridge, UK). Subsequently, the sections were incubated at 37 °C for 50 min with FITC-conjugated secondary antibodies: goat anti-mouse IgG (ab6785, Abcam, Cambridge, UK) and goat anti-rabbit IgG (ab6717, Abcam, Cambridge, UK). After nuclear counterstaining with DAPI for 10 min, the fluorescent images were captured utilizing a fluorescence microscope. Signal intensity was quantified with ImageJ software ($n = 3$ per group).

2.15. Statistical analysis

Data analysis was performed with SPSS version 26.0 (IBM, Inc., Chicago, USA). Values are presented as mean $\pm$ standard deviation (SD) with exceptions explicitly stated. Comparisons between two groups were conducted by independent samples *t*-test. Intergroup comparisons were analyzed by one-way analysis of variance (ANOVA) with LSD post hoc testing when parametric assumptions (normality and homoscedasticity) were satisfied. If these assumptions were not satisfied, the Kruskal-Wallis H test was used instead. Statistical significance was defined as $P < 0.05$.

3. Results

3.1. UPLC-Q-TOF/MSE analysis of JQHF

JQHF consists of 9 herbal ingredients (Table 1). The chemical profile of JQHF was analyzed using UPLC-Q-TOF/MSE. The representative chromatogram is presented in Fig. S1. As shown in Table S1, a total of 28 components were identified across the 9 herbal ingredients. The identification was built upon retention times and UV spectra, confirmed through contrast with authentic reference standards.

3.2. NAC inhibits sC5b-9-induced ROS production, mitochondrial dysfunction and podocyte damage in vitro

ROS serve as markers of mitochondrial dysfunction and damage. To investigate this, we established an in vitro model of podocyte injury induced by sublytic C5b-9 (sC5b-9). N-acetylcysteine (NAC), a known ROS inhibitor, and palmitic acid (PA), a ROS inducer, were employed to evaluate their respective effects. sC5b-9 stimulation remarkably increased intracellular ROS levels, as measured by DCFH-DA fluorescence staining using a fluorescence microplate reader. NAC effectively inhibited ROS production in podocytes, whereas PA markedly enhanced ROS accumulation (Fig. 1C). The decrease of MMP level is one of the indicators reflecting mitochondrial dysfunction. sC5b-9 exposure resulted in disrupted MMP, similar to the PA group, while NAC treatment significantly restored MMP levels (Fig. 1A and D). TEM revealed pronounced mitochondrial ultrastructural damage in both the model and PA groups. In contrast, NAC treatment notably ameliorated these mitochondrial abnormalities (Fig. 1B).

Furthermore, WB analysis was conducted to detect podocyte-specific markers. The expression of podocin, a protective protein critical for podocyte integrity, was remarkably downregulated of model group in comparison with control group. NAC intervention restored podocin levels, whereas PA further suppressed its expression. Conversely, desmin, a marker of podocyte injury, was upregulated in the model and PA groups, with NAC treatment significantly reducing its expression (Fig. 1E and F). Collectively, these results suggested that there were ROS accumulation, mitochondrial dysfunction and podocytes injury in our model. The injured podocytes possibly could be ameliorated by eliminating mitochondrial ROS and modulating mitochondrial function.

3.3. NAC inhibits sC5b-9-induced NLRP3 activation in vitro

To examine the relationship between mitochondrial dysfunction and inflammatory factor activation in podocytes, we assessed the impact of NAC and PA on sC5b-9-induced NLRP3 inflammasome activation. Immunofluorescence analysis revealed that sC5b-9 stimulation significantly increased the expression of NLRP3, caspase-1, IL-1 β , and IL-6 in podocytes. Intervention with NAC markedly suppressed the expression of NLRP3 and caspase-1, along with reduced the levels of the pro-inflammatory cytokines IL-1 β and IL-6. On the contrary, PA intervention intensified the expression of these inflammatory markers, indicating enhanced activation of inflammasomes (Fig. 2A and B). The above results were further confirmed by WB detection (Fig. 2C and D).

Collectively, these results demonstrate that ROS inhibition effectively attenuates complement-mediated activation of NLRP3 inflammasome and release of related inflammatory factors in podocytes.

3.4. JQHF improved the renal damage of PHN rats

We observed that the PHN model rats exhibited significantly elevated levels of total cholesterol (TC) and 24-h urinary protein (24h UTP), along with a marked reduction in serum albumin (ALB) compared with the normal group (Fig. 3A–C). After six weeks of treatment, these biochemical abnormalities were significantly ameliorated in the JQHF-M group. Notably, JQHF-M outperformed benazepril in therapeutic effectiveness. Similarly, JQHF-H treatment significantly reduced 24h UTP and improved ALB levels compared to the model group. However, unlike JQHF-M, JQHF-H treatment did not produce significant reduction in TC levels. Importantly, the combination of JQHF-M with the 3-MA led to a substantial increase in 24h UTP and TC levels and a notable reduction in ALB in comparison with the JQHF-M group. These findings demonstrate that suppressing autophagy can affect the efficacy of JQHF.

The pathological morphology of kidney tissue and immunofluorescent staining of IgG and C5b-9 were observed utilizing a light microscope (Fig. 3D). The rats of model group exhibited characteristic pathological features of IMN, including mesangial cell proliferation, thickening of the glomerular basement membrane, tubular protein casts, focal glomerulosclerosis, and prominent IgG and C5b-9 deposition along the glomerular capillary walls. Both JQHF-M and JQHF-H treatments significantly attenuated these renal pathological alterations, showing greater improvements than the benazepril group (Fig. 3E). Additionally, JQHF-M markedly reduced IgG and C5b-9 deposition in comparison to the model group (Fig. 3F). Notably, the renoprotective effects of JQHF were significantly diminished in the JQHF-M+3 MA group, as evidenced by increased tubular damage and elevated IgG and C5b-9 deposition compared to the JQHF-M group. All of the results further support the hypothesis that autophagy inhibition compromises the efficacy of JQHF in PHN rats.

3.5. JQHF protects against accumulation of mtROS in the kidney of PHN rats

As illustrated in Fig. 4A, B, 4D along with 4E, in comparison to the normal group, the level of ROS was markedly elevated and MMP was decreased in PHN rats. After six weeks of treatment, JQHF-M as well as JQHF-H group revealed a significant reduction in ROS levels and a substantial improvement in MMP contrasted with the model group. Similarly, the benazepril group demonstrated a reduction in ROS levels; however, the improvement in MMP was not statistically significant in comparison with the model cohort.

TEM was utilized to evaluate mitochondrial morphology in PHN rat renal tissue. Control animals displayed mitochondria with preserved double-membrane architecture and distinct, well-aligned cristae. In contrast, mitochondria from the model group showed pronounced swelling, vacuolization, and disrupted cristae. Notably, treatment with JQHF-M and JQHF-H significantly alleviated these ultrastructural abnormalities, with improved mitochondrial integrity compared to both the model and benazepril groups (Fig. 4C). Notably, compared with JQHF-M group, the mitochondrial dysfunction, and the expressions of ROS in the JQHF-M+3 MA group were significantly increased. Experimental results confirm that autophagy inhibition compromises the mitochondrial protective effects of JQHF, highlighting the critical role of autophagy in mediating its therapeutic efficacy.

3.6. JQHF inhibits the activation of NLRP3 corpuscles in the kidney of PHN rats

As shown in Fig. 5C, F, and 5G, the expression of NLRP3 and Caspase-

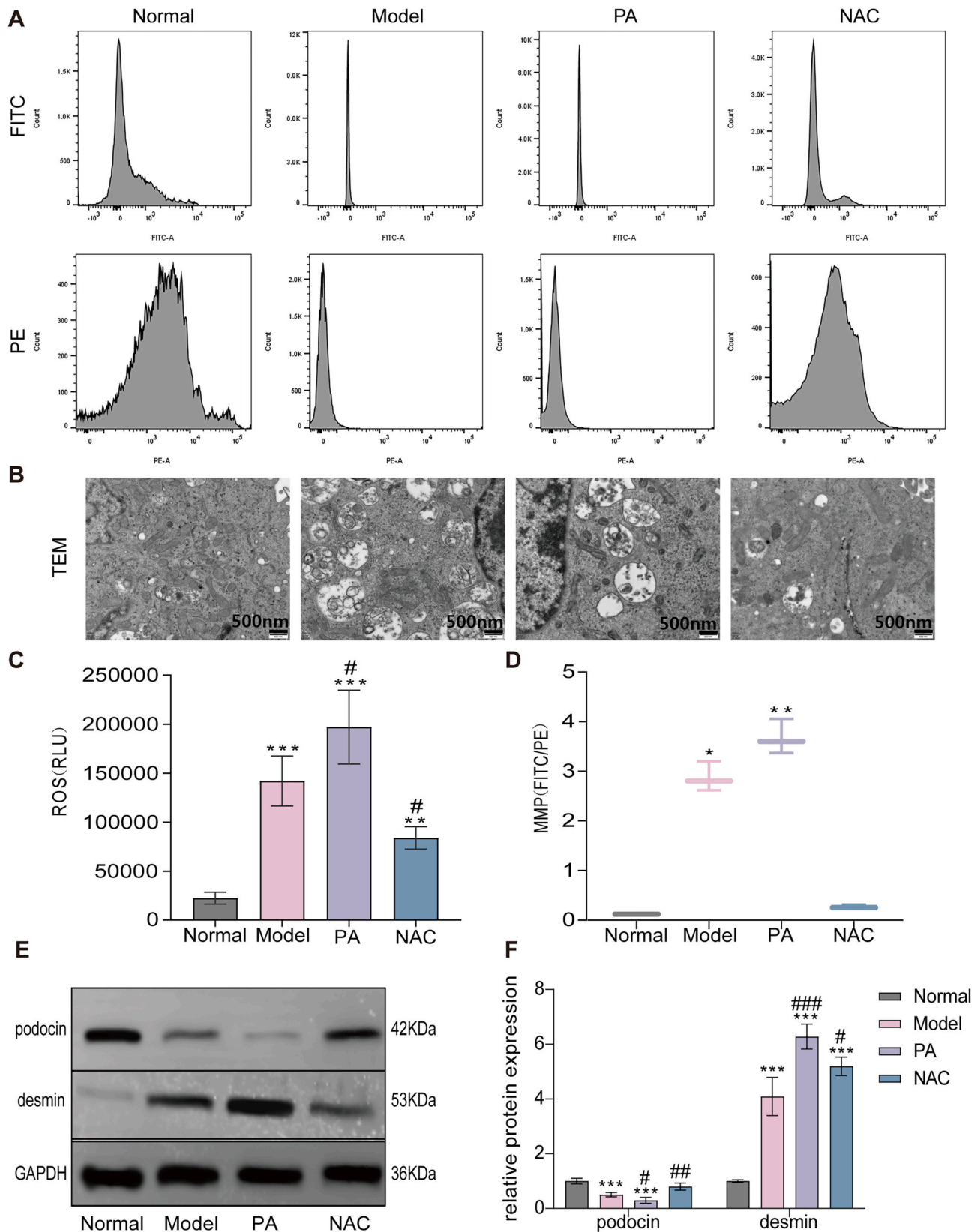


Fig. 1. NAC inhibited sC5b-9-induced ROS production and podocytes injury in vitro.

(A) Representative images of MMP detected by flow cytometry ($n = 3$). (B) Mitochondrial morphology was observed by TEM ($n = 3$). Scale bars = 500 nm. (C) Quantitative analysis of ROS assessed by fluorescence enzyme immunoassay instrument ($n = 3$). (D) Relative ratios of JC-1 FITC/PE fluorescence intensity were assessed by flow cytometry to verify the MMP in podocytes ($n = 3$). (E, F) Representative immunoblot images and quantitative analysis of podocin and desmin measured by Western blotting ($n = 3$). MMP, mitochondrial membrane potential; TEM, transmission electron microscopy; ROS, reactive oxygen species; PA, palmitic acid, a ROS inducer; NAC, N-Acetylcysteine, a ROS scavenger. $^*P < 0.05$, $^{**}P < 0.01$, $^{***}P < 0.001$ vs. Normal group; $^{\#}P < 0.05$, $^{##}P < 0.01$, $^{###}P < 0.001$ vs. Model group.

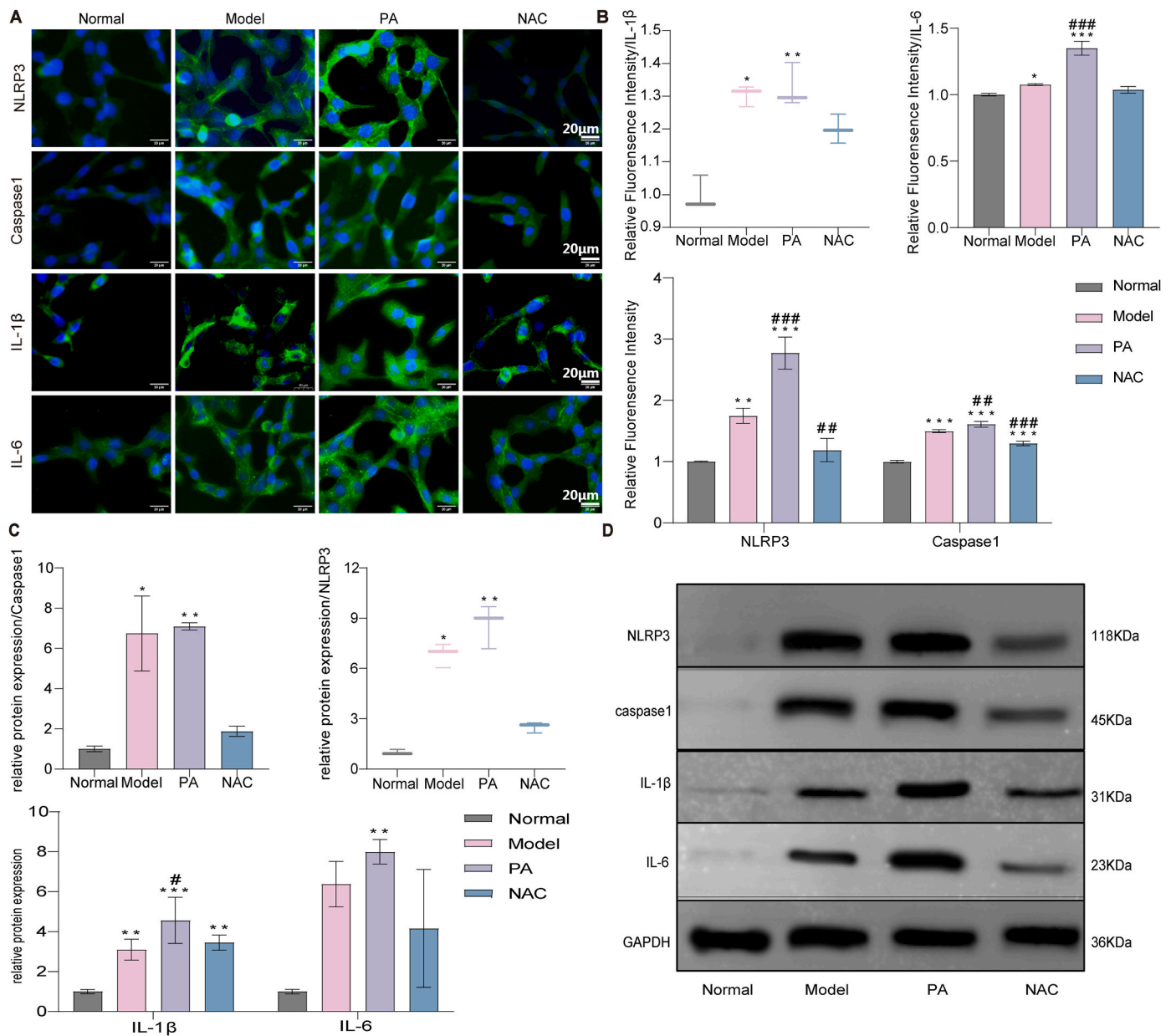


Fig. 2. NAC inhibited sC5b-9-induced NLRP3 activation in vitro.

(A, B) Representative immunofluorescent images and quantitative analysis of fluorescence intensity of NLRP3, Caspase-1, IL-1 β and IL-6 measured by fluorescence ($n = 3$). Scale bar = 20 μ m. (C, D) Representative immunoblot images and quantitative analysis of NLRP3, Caspase-1, IL-1 β and IL-6 measured by Western blotting ($n = 3$). PA, palmitic acid, a ROS inducer; NAC, N-Acetylcysteine, a ROS scavenger. * $P < 0.05$, ** $P < 0.01$, *** $P < 0.001$ vs. Normal group; # $P < 0.05$, ## $P < 0.01$, ### $P < 0.001$ vs. Model group.

1 were remarkably elevated in PHN model rats versus normal controls, indicating inflammasome activation. Six-week JQHF-M treatment substantially attenuated these elevations, demonstrating the anti-inflammatory effect of JQHF.

Furthermore, podocytes, highly specialized epithelial cells lining the glomerular capillary tuft, are critical for sustaining the intactness of the glomerular filtration barrier. Podocytes injury together with depletion are characteristics of IMN. To evaluate podocyte health, we assessed the level of the podocyte-specific protein Nephlin using WB analysis and immunohistochemistry. As shown in Fig. 5B and E, WB analysis showed that the expression of Nephlin was markedly augmented in the Benazepril, JQHF-M along with JQHF-H groups by comparison to the model group. Notably, both JQHF-M and JQHF-H groups exhibited higher Nephlin expression levels than the Benazepril group, suggesting superior podocyte-protective effects. Consistent with these findings,

immunohistochemical staining (Fig. 5A–D) showed that JQHF-M significantly upregulated Nephlin expression in renal tissues in comparison with the model cohort.

Importantly, the JQHF-M+3MA group demonstrated significantly elevated Caspase-1 expression but reduced Nephlin levels relative to the JQHF-M group. The above findings suggest that autophagy inhibition compromises the protective effects of JQHF, underscoring the critical role of autophagy in mediating its therapeutic efficacy.

3.7. JQHF upregulates the PINK1 pathway to exert renoprotective effects in vitro

Since PINK1 is a key regulator of autophagy, we investigated whether JQHF exerts its renoprotective effects through the PINK1 pathway in vitro. In comparison to the normal group, ROS accumulation

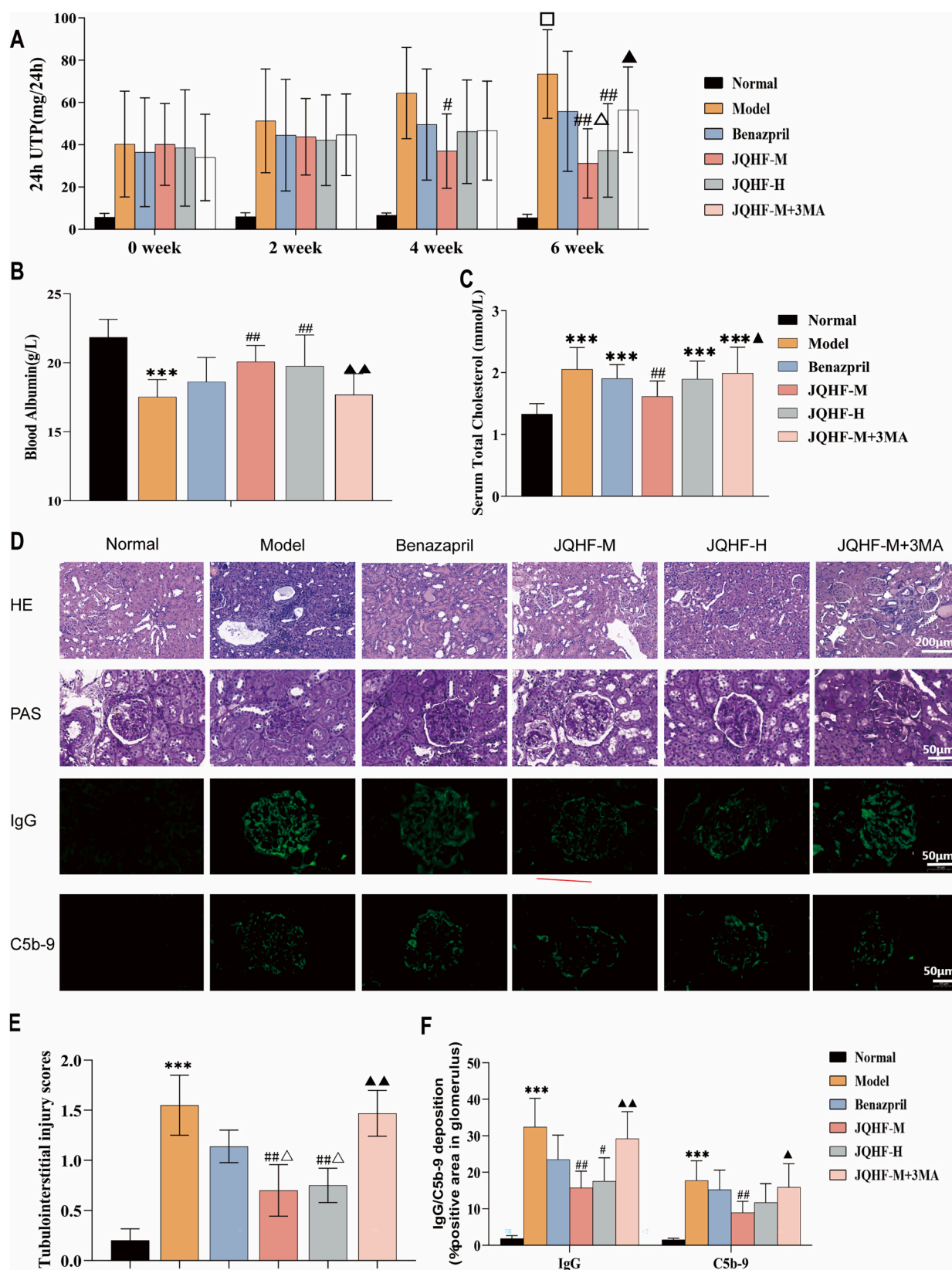


Fig. 3. Effects of JQHF on biochemical parameters and renal pathology in PHN rats.

(A) 24h urinary protein (n = 7/8). (B) ALB (n = 7/8). (C) TC (n = 7/8). (D) Representative microstructural images of different groups (n = 6). HE staining, scale bar = 200 μm. PAS, C5b-9 and IgG, scale bar = 50 μm. (E) Tubulointerstitial injury scores (n = 6). (F) The positive area of C5b-9 and IgG deposition in glomerulus (n = 6). ****P* < 0.001 vs. Normal group; #*P* < 0.05 vs. Model group, ##*P* < 0.01 vs. Model group; △*P* < 0.05 vs. Benazepril group; ▲*P* < 0.05 vs. JQHF-M group, ▲▲*P* < 0.01 vs. JQHF-M group; Compared with before treatment □ *P* < 0.05.

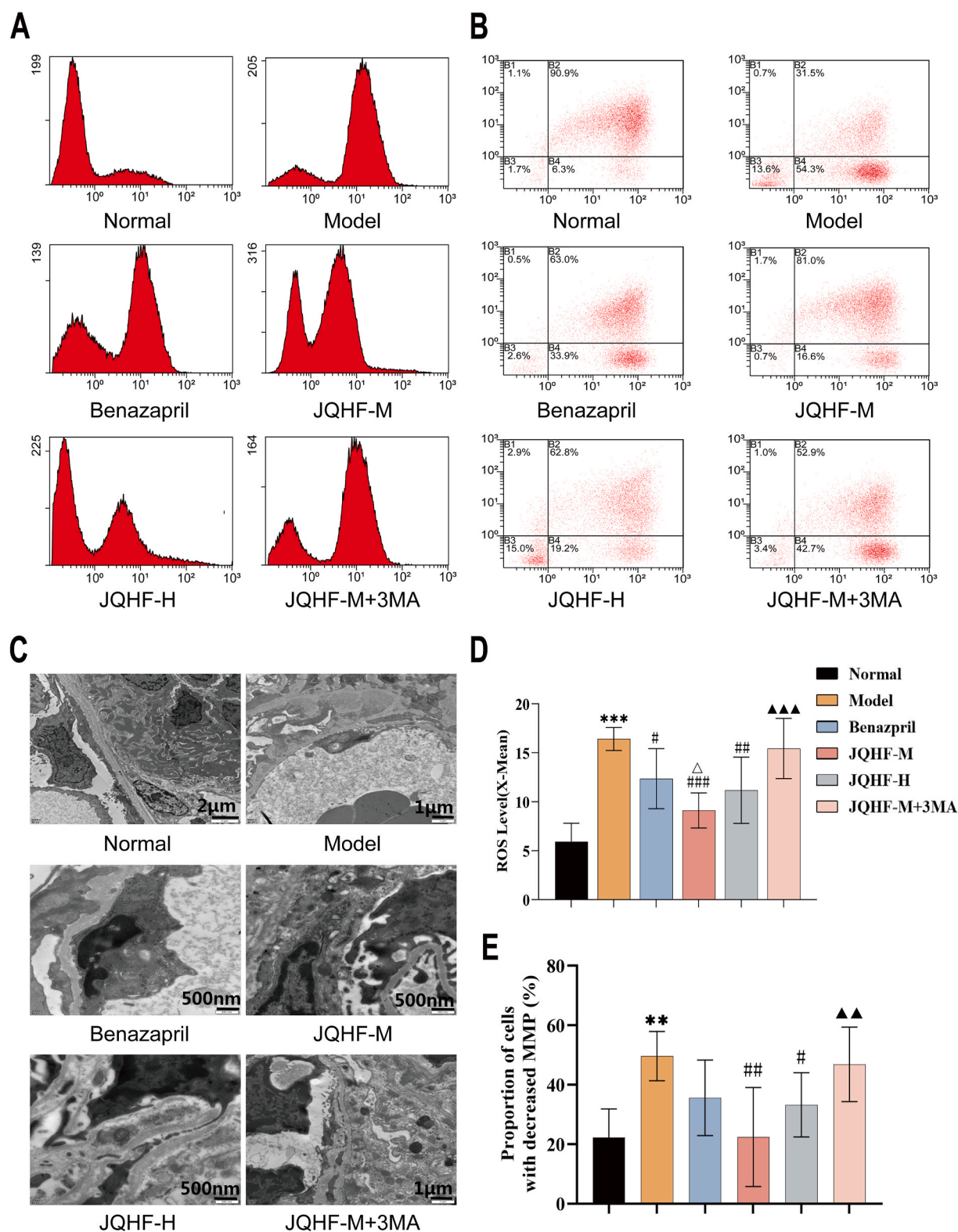


Fig. 4. Effects of JQHF on against ROS-induced mitochondrial dysfunction in PHN rats.

(A) Determination ROS production by DCFA staining (n = 6). (B) Determination of MMP by JC-1 staining (n = 6). (C) Analyze mitochondrial ultrastructural injuries by TEM (n = 6). (D) Quantitative analysis of intracellular ROS level (n = 6). (E) Quantitative analysis of MMP (n = 6). ** $P < 0.01$ vs. Normal group, *** $P < 0.001$ vs. Normal group; # $P < 0.05$ vs. Model group, ## $P < 0.01$ vs. Model group, ### $P < 0.001$ vs. Model group; △ $P < 0.05$ vs. Benazapril group; ▲▲ $P < 0.01$ vs. JQHF-M group, ▲▲▲ $P < 0.001$ vs. JQHF-M group.

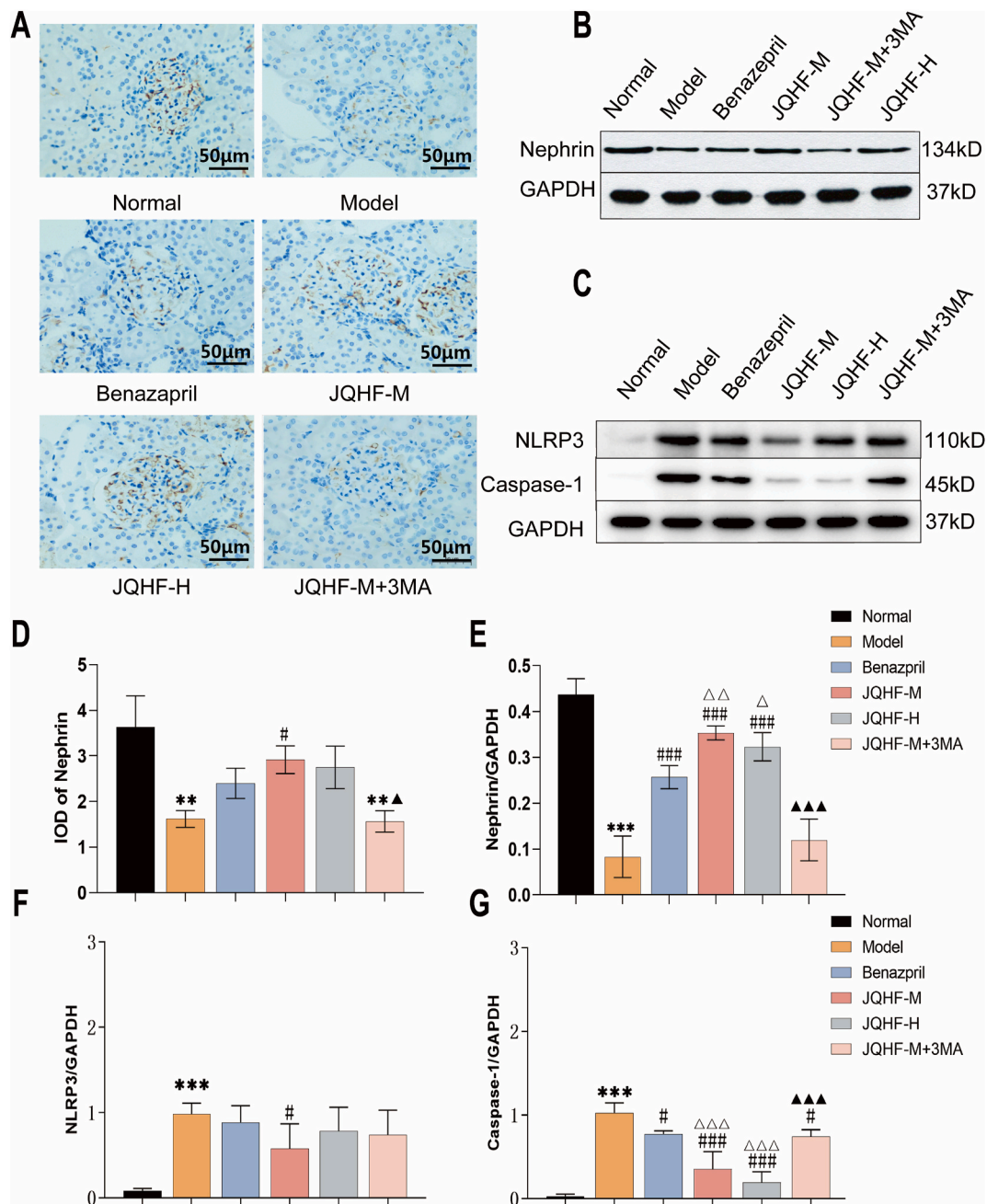


Fig. 5. JQHF inhibits NLRP3 inflammasome activation and improves podocyte injury in the kidneys of PHN rats.

(A) Immunohistochemistry staining of Nephlin protein (n = 6). Scale bar = 50 μ m. (B, C) Western blot analysis of the expression of Nephlin, caspase-1 and NLRP3 protein in each groups (n = 6). (D) Analysis of IOD of Nephlin (n = 6). (E–G) Quantitative analysis of the expression of Nephlin, caspase-1 and NLRP3 protein (n = 6). ** P < 0.01 vs. Normal group, *** P < 0.001 vs. Normal group; # P < 0.05 vs. Model group, ### P < 0.001 vs. Model group; Δ P < 0.05 vs. Benazepril group, $\Delta\Delta$ P < 0.01 vs. Benazepril group, $\Delta\Delta\Delta$ P < 0.001 vs. Benazepril group; $\blacktriangle$ P < 0.05 vs. JQHF-M group, $\blacktriangle\blacktriangle\blacktriangle$ P < 0.001 vs. JQHF-M group.

was sharply augmented in both the model group and the shRNA-PINK1 group. Intervention with JQHF markedly suppressed ROS production; however, the protective effect of JQHF was significantly attenuated when co-administered with shRNA-PINK1 (Fig. 6C). To further assess mitochondrial function, we measured the MMP. As shown in Fig. 6A and D, MMP was significantly disrupted in the model and shRNA-PINK1 groups. JQHF treatment restored MMP levels. TEM was employed to observe mitochondrial ultrastructural changes in podocytes across different groups. The model group exhibited substantial mitochondrial damage. JQHF treatment significantly improved mitochondrial ultrastructure compared to the model group (Fig. 6B).

Further, podocyte markers podocin and desmin were subsequently

evaluated. Our results showed that level of podocin protein, was simultaneously down-regulated in the group of model, shRNA-PINK1 and JQHF co-application of the shRNA-PINK1 relative to the normal group. JQHF intervention markedly upregulated podocin levels versus the model group, an effect attenuated by PINK1 silencing, indicating PINK1-mediated podocyte protection. Conversely, desmin, a marker of podocyte injury, was overexpressed in the model group. JQHF effectively reduced desmin expression, while co-treatment with shRNA-PINK1 only partially reversed desmin levels, indicating an incomplete protective response without functional PINK1 (Fig. 6E and F). Collectively, these findings suggest that JQHF mitigates ROS accumulation, enhances mitochondrial function, and protects against podocyte injury

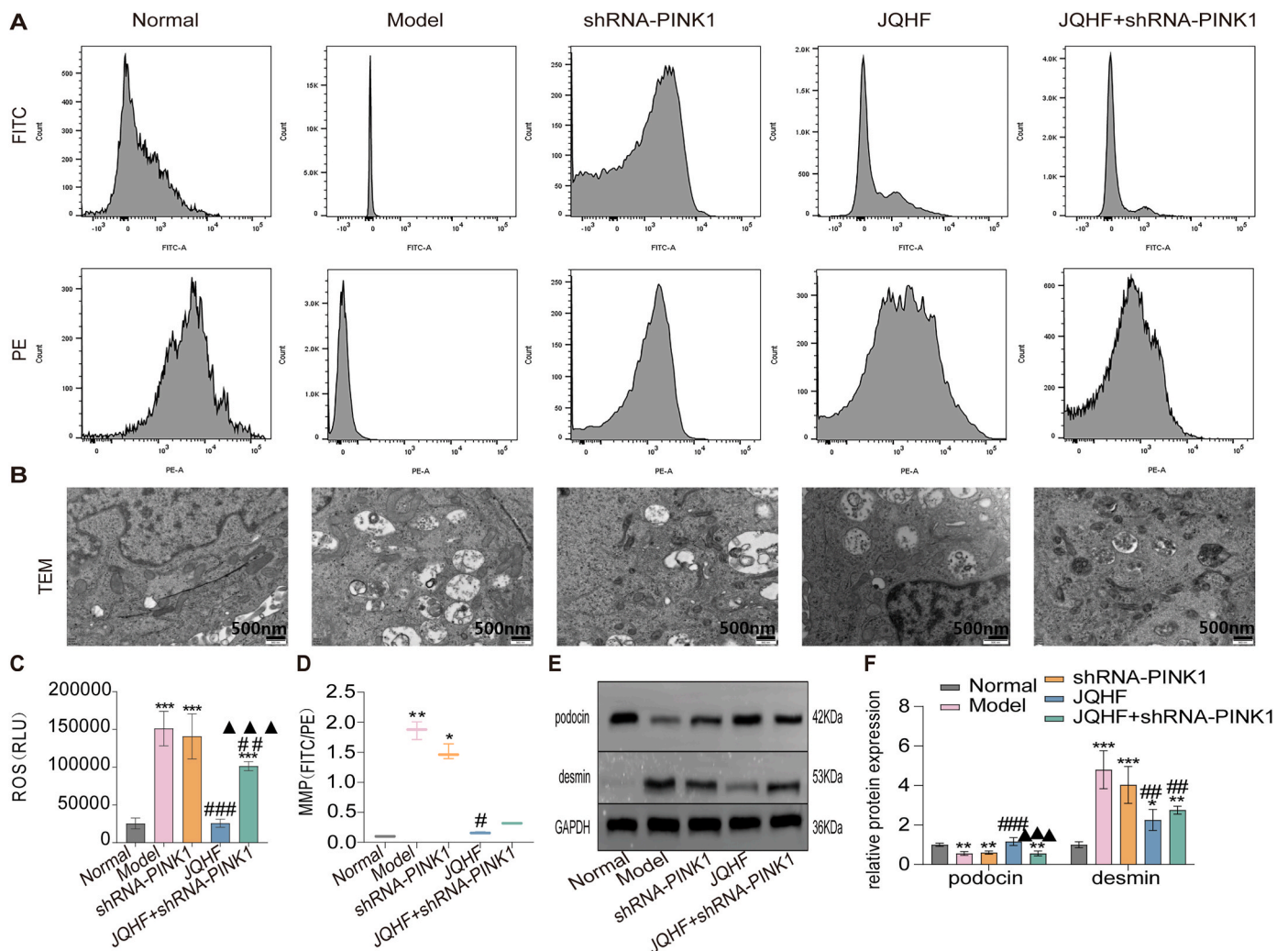


Fig. 6. JQHF inhibited sC5b-9-induced ROS production and podocytes injury through upregulating the PINK1 pathway in vitro. (A) Representative images of MMP detected by flow cytometry ($n = 3$). (B) Mitochondrial morphology was observed by TEM ($n = 3$). Scale bars = 500 nm. (C) Quantitative analysis of ROS assessed by fluorescence enzyme immunoassay instrument ($n = 3$). (D) Relative ratios of JC-1 FITC/PE fluorescence intensity were assessed by flow cytometry to verify the MMP in podocytes ($n = 3$). (E, F) Representative immunoblot images and quantitative analysis of podocin and desmin measured by Western blotting ($n = 3$). MMP, mitochondrial membrane potential; TEM, transmission electron microscopy; ROS, reactive oxygen species; JQHF, Jianpi Qushi Heluo Formula; JQHF + shRNA-PINK1, application of JQHF therapy on the basis of silencing PINK1. * $P < 0.05$, ** $P < 0.01$, *** $P < 0.001$ vs. Normal group; # $P < 0.05$, ## $P < 0.01$, ### $P < 0.001$ vs. Model group; ▲▲ $P < 0.01$ vs. JQHF group.

primarily through the upregulation of the PINK1 pathway.

3.8. JQHF inhibits sC5b-9-induced NLRP3 activation through upregulating the PINK1 pathway in vitro

To clarify the effect of JQHF in regulating inflammatory responses, we further investigated the activation of the NLRP3 corpuscles and associated inflammatory factors in podocytes using immunofluorescence analysis. As depicted in Fig. 7A, C, and 7D, a pronounced elevation in NLRP3 and IL-1 β was observed in the model group, shRNA-PINK1 group along with JQHF + shRNA-PINK1 group in contrast to the normal group. Intervention with JQHF markedly suppressed the over-expression of NLRP3 and IL-1 β ; however, this inhibitory effect was significantly attenuated when PINK1 was silenced. Consistent with these findings, similar results were observed of NLRP3 through WB testing (Fig. 7E–G). Additionally, we measured the level of Caspase-1 and IL-6 via immunofluorescence. As depicted in Fig. 7A, B, and 7D, the model and PINK1-silenced groups showed significant increases in Caspase-1 and IL-6. JQHF intervention effectively downregulated their expression. However, the JQHF + shRNA-PINK1 group exhibited elevated

Caspase-1 and IL-6 levels relative to the JQHF group, though this trend lacked statistical significance. Simultaneously, no significant differences in Caspase-1, IL-6, or IL-1 β expression were detected by WB analysis between JQHF and JQHF + shRNA-PINK1 groups (Fig. 7E–H). Overall, the experimental outcomes confirm that JQHF can inhibit NLRP3 inflammasome activation and downstream inflammatory responses in podocytes by enhancing PINK1-mediated mitophagy.

Generally speaking, all of findings establish PINK1-mediated mitophagy as a crucial pathway in IMN pathogenesis. The upregulating of PINK1 by JQHF may be conducive to improving mitochondrial dysfunction and inhibiting the activation of NLRP3 inflammatory vesicles, which in turn attenuates podocytes injury and local inflammatory response, and exerts renoprotective effects.

4. Discussion

In this research, we elucidated the renal protective effects of JQHF in relation to podocyte mitochondrial function and its underlying mechanisms. Our findings demonstrate that JQHF significantly increases serum ALB, reduces 24h UTP and TC levels in PHN rats. Mechanistically,

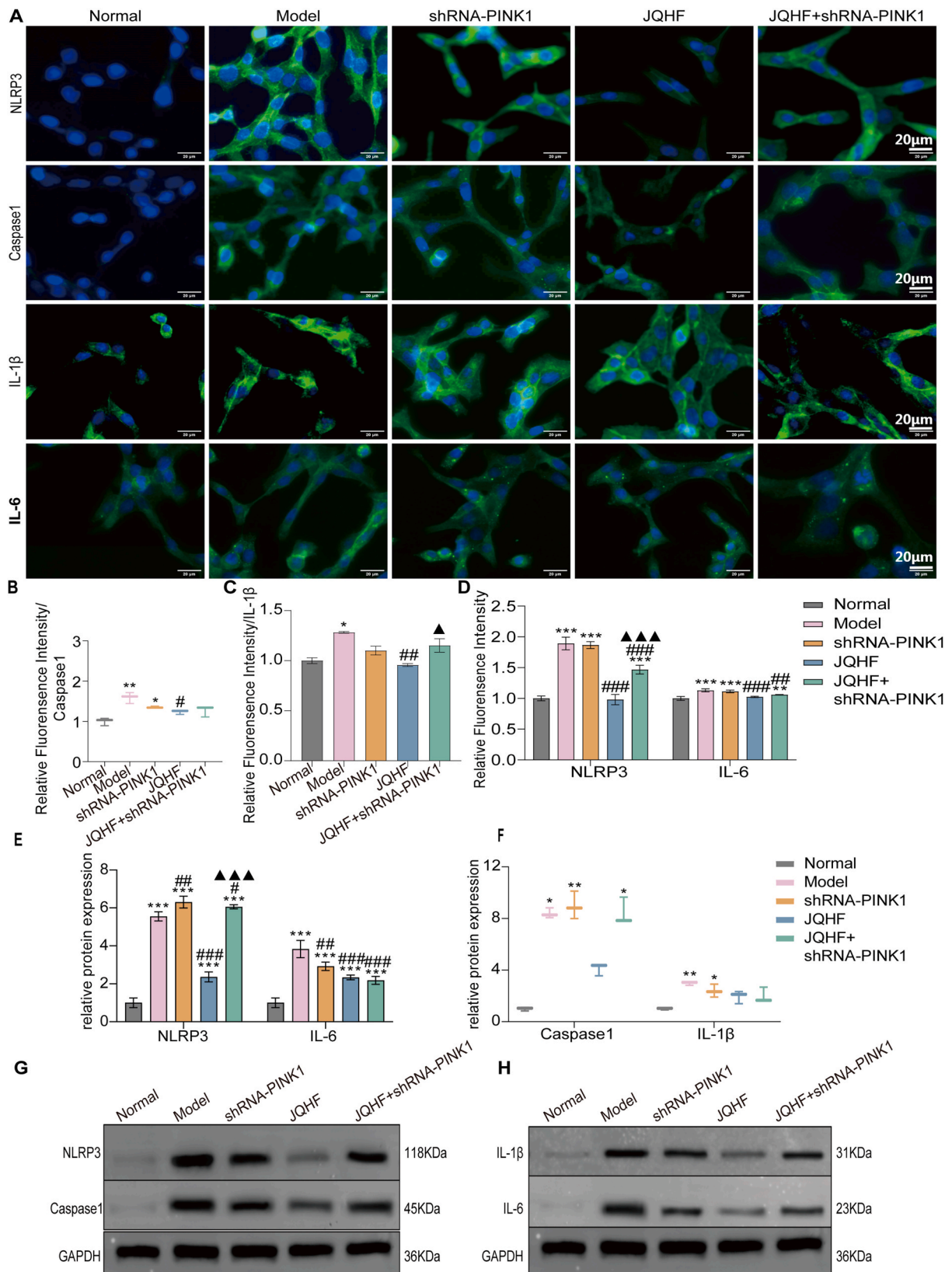


Fig. 7. JQHF inhibited sC5b-9-induced NLRP3 activation through upregulating the PINK1 pathway in vitro. (A–D) Representative immunofluorescent images and quantitative analysis of fluorescence intensity of NLRP3, Caspase-1, IL-1 β and IL-6 measured by fluorescence ($n = 3$). Scale bar = 20 μ m. (E–H) Representative immunoblot images and quantitative analysis of NLRP3, Caspase-1, IL-1 β and IL-6 measured by Western blotting ($n = 3$). JQHF, Jianpi Qushi Heluo Formula; JQHF + shRNA-PINK1, application of JQHF therapy on the basis of silencing PINK1. * $P < 0.05$, ** $P < 0.01$, *** $P < 0.001$ vs. Normal group; # $P < 0.05$, ## $P < 0.01$, ### $P < 0.001$ vs. Model group; ▲ $P < 0.05$, ▲▲ $P < 0.01$ vs. JQHF group.

these protective effects are mediated through the regulation of PINK1-dependent mitophagy, which suppresses ROS-mediated NLRP3 inflammasome activation, thereby attenuating podocyte injury.

The NLRP3 inflammasome emerges as a key regulator of inflammation. Upon activation, it triggers caspase-1, provoking the maturation and release of pro-inflammatory cytokines including IL-1 β (Zhao et al., 2020). NLRP3 is a well-established target for managing metabolic and inflammatory diseases. Previous studies have identified NLRP3 inflammasome activation was strongly related to podocyte injury. For example, suppression of NLRP3 inflammasome ameliorated podocyte damage in diabetic nephropathy, by IL-1 β /ROS/NF- κ B p65 mediated diabetes-related excessive lipid droplet formation (Wu et al., 2021). Moreover, in hyperglycemic mouse models, podocyte-specific NLRP3 gain-of-function mutations exacerbated kidney damage, while NLRP3 deficiency conferred renal protection (Shahzad et al., 2022). The role of NLRP3 also received attention in IMN. A study found that overexpression of NLRP3, IL-1 β , and caspase-1 has been observed both in vivo and in vitro. Here, all sorts of injury as reflected in local inflammation of renal tissue, podocytes apoptosis and over expression of desmin were reduced, in contrast, the expression of podocytes specific structural proteins Nephritin and Podocin were increased by inhibiting the activation of NLRP3 inflammasome (Liu et al., 2019). Consistent with these findings, our study demonstrated elevated NLRP3 and caspase-1 levels alongside podocyte injury markers in IMN models, confirming that NLRP3 inflammasome is activated in IMN and may ultimately lead to podocytes injury and disease progression.

Presently, It is widely accepted that NLRP3 inflammasome activation stems from diverse upstream signals, notably K⁺ efflux, ROS production, and mitochondrial ETC dysfunction (Billingham et al., 2022; Youm et al., 2015). Among them, elevated ROS triggers NLRP3 inflammasome activation by impairing mitochondrial function, leading to inflammatory cascades, which has been confirmed in various diseases (Feng et al., 2017; Jiang et al., 2024; Traba et al., 2015; Wang et al., 2024). For instance, the introduction of ROS inhibitor NAC could significantly decrease the accumulation of ROS, thus reducing the expression of NLRP3, and correspondingly reducing the level of inflammatory factors such as IL-1 β and IL-6 in intestinal inflammation models (Shan et al., 2024; Xu et al., 2023). Nevertheless, the correlation between ROS and NLRP3 inflammasome activation remains unknown in the IMN. Hence, the effect of ROS on mediating activation of NLRP3 inflammasome was evaluated by simultaneously using ROS inhibitor and inducer, which namely NAC and PA in vitro. Our research results showed that the sC5b-9 induced podocytes model was challenged with significantly accumulated ROS and decreased MMP, accompanied by overexpression of NLRP3 protein, low expression of podocin protein, and high expression of desmin protein. Intriguingly, PA could exacerbate excessive ROS production, develop a vigorous overexpression of NLRP3, as did some related inflammatory proteins, also worsen podocyte damage, while the results were reversed in response to NAC. Our results provide a unifying mechanism that ROS mediates the activation of NLRP3 inflammasome in the podocytes of IMN. Eliminating excess ROS emerges as a promising strategy to mitigate inflammasome activation and podocyte damage.

JQHF, first developed by the Nephrology Department of Xiyuan Hospital, has shown clinical efficacy in IMN, though its precise mechanisms remained elusive. In this context, we attempted to clarify the effects of JQHF on mitochondrial dysfunction, specifically on ROS generation and subsequent NLRP3 inflammasome activation. Our tests have shown that mitochondrial damage and ROS overproduction in podocytes can be suppressed by JQHF treatment. In addition, JQHF can significantly inhibit the NLRP3 inflammasome activation as reflected by low expression of NLRP3, Caspase-1, IL-1 β , and IL-6. Furthermore, JQHF treatment can significantly upregulate the expression of podocin, a cytoprotective protein in podocytes, while concurrently downregulating the overexpression of desmin, a marker associated with podocyte injury. These effects collectively attenuated podocyte damage in both in vitro and in vivo models.

JQHF comprises nine traditional Chinese medicinal herbs (Table 1), with components known to target oxidative stress, mitochondrial dysfunction, and inflammation via diverse signaling pathways. For instance, astragaloside IV, derived from *Astragalus membranaceus* (Fisch.) Bge., downregulates NLRP3, ASC, and caspase-1 expression through Nrf2-ARE/TFAM and SIRT1/PGC1 α /Nrf1 pathways, ameliorating inflammation (Li et al., 2024; Shen et al., 2023; Sun et al., 2024). Formononetin is also one of the components identified in *Astragalus membranaceus* (Fisch.) Bge. Previous studies have demonstrated that formononetin attenuates hepatic ischemia-reperfusion (I/R) injury in rat models by activating PINK1/PARKIN-mediated mitophagy. Notably, genetic knockdown of PINK1 abolished the hepatoprotective effects of formononetin, suggesting the essential role of this pathway in its therapeutic mechanism (Ma et al., 2022). Tetrandrine, a component identified in *Stephania tetrandra* S.Moore, exhibits potent anti-inflammatory properties through restraining NLRP3 corpuscle activation (Song et al., 2022). Magnoflorine is also one of the components identified in *Stephania tetrandra* S.Moore. As previously reported, Magnoflorine can improve kidney damage in mice fed with lipogenic and gluconeogenic diets by facilitating Parkin/PINK1 dependent mitophagy and inhibiting NLRP3 mediated pyroptosis (Cheng et al., 2024). Diosgenin, derived from *Dioscorea nipponica* Makino, can effectively reduce blood glucose levels and ameliorate renal pathological damage in diabetic nephropathy rats. The underlying mechanism involves diosgenin's ability to regulate mitochondrial dynamics and enhance mitophagy through the PINK1-MFN2-Parkin pathway (Zhong et al., 2023). Additionally, L-Arginine is one of the main components identified in *Poria cocos* (Schw.) Wolf. The prior investigations have proved that L-Arginine can significantly improve the inflammatory response during the cardiovascular and pulmonary diseases, which related to signaling pathways including ROS/NLRP3/NF- κ B and SIRT1 (Ma et al., 2023; Zhang et al., 2021). On the whole, the mechanisms of action of these identified components are largely consistent with the focus of JQHF in this study. JQHF may be an effective strategy for treating IMN, its mechanism of therapeutic involves mitochondrial dysfunction and NLRP3 inflammasome activation.

PINK1/Parkin-mediated mitophagy exerts a pivotal role in maintaining mitochondrial quality by removing damaged mitochondria. Upon mitochondria damage, abnormal mitochondrial signals are detected by PINK1, which stable aggregation on the cytoplasmic surface of external mitochondrial bilayer, then gave rise to mobilization and triggering of Parkin and converts it to an activated E3 ubiquitin ligase. Followed by initiating multiple signaling events and continuously degrading defective mitochondria to maintain an appropriate mitochondrial population in the cell (Lazarou et al., 2015). A substantial body of researches have confirmed that mitophagy deficiency is a potential mechanism for disease pathogenesis. By upregulating the PINK/Parkin pathway, not only can the overexpression of mtROS be reduced, but it is also an effective way to inhibit triggering of NLRP3 inflammasome. The prior study found that the expression of PINK and Parkin was reduced, accompanied by mitochondrial phagocytosis defect, and ROS accumulation of diabetes mice (Xiao et al., 2017). Besides, mtROS and subsequent NLRP3 inflammasome activation were limited through mitophagy mediated by PINK1/Parkin to inhibit cell apoptosis and tissue damage in models of acute kidney injury both in cell-based and animal models (Lin et al., 2019). Similarly, hesperetin can inhibit mitochondrial damage in HepG2 cells by participating in the PINK1/Parkin mediated mitophagy pathway, which could decrease excessive accumulation of mtROS and improve decreasing in MMP, and alleviate PA triggered activation of NLRP3 inflammasomes and cell death in metabolism-associated fatty liver disease (Li et al., 2024).

Considering the pivotal role of mitophagy in sustaining mitochondrial homeostasis and modulating activation of NLRP3 inflammasome, we further tested the hypothesis that the protective effects of JQHF are compromised when the PINK1/Parkin-mediated mitophagy pathway is inhibited. To validate this, we established a podocyte model with PINK1

silencing. Our findings revealed that while JQHF significantly reduced ROS production and suppressed NLRP3 expression, its protective effects on podocytes were markedly attenuated in the absence of PINK1. Overall, these results established that regulating mitophagy based on the PINK1/Parkin pathway may be a promising mechanism for JQHF to improve mitochondrial function, inhibit inflammatory response against podocytes injury.

Furthermore, NLRP3 inflammasome activation induces pro-caspase-1 cleavage, generating active caspase-1 that mediates IL-1 β and IL-18 maturation. Concurrently, activated caspase-1 processes GSDMD, liberating its N-terminal domain to form membrane pores—a mechanism driving inflammogenic factor release and pyroptotic cell death (Huang et al., 2021). Overall, our research mainly focuses on JQHF limiting ROS release through the mitochondrial pathway, thus repressing NLRP3 inflammasome activation. However, in our podocyte injury model, compelling evidence supporting NLRP3 inflammasome-mediated pyroptosis remains elusive. The potential of JQHF to modulate pyroptosis by targeting the NLRP3 inflammasome represents an important avenue for future research.

5. Conclusion

Collectively, our study demonstrates that JQHF is a promising therapeutic strategy for treating IMN. By upregulating the PINK1/Parkin mitophagy pathway, JQHF effectively inhibits ROS-mediated activation of the NLRP3 inflammasome, thus alleviating podocyte injury. These results contribute to a better understanding of the mechanisms underlying JQHF's therapeutic efficacy and highlight potential avenues for future research in IMN treatment.

CRediT authorship contribution statement

Lei Yan: Writing – original draft, Investigation. **Qin Zeng:** Writing – original draft, Formal analysis. **Wenru Wang:** Visualization, Data curation. **Ying Liang:** Formal analysis, Data curation. **Rui Lang:** Investigation, Data curation. **Jingyi Zhan:** Methodology, Data curation. **Jiayi Yang:** Validation. **Renhuan Yu:** Supervision, Methodology, Funding acquisition, Conceptualization. **Xinhui Wang:** Writing – review & editing, Supervision, Funding acquisition, Conceptualization.

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Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Appendix A Supplementary data

Supplementary data to this article can be found online at <https://doi.org/10.1016/j.jep.2025.120291>.

Data availability

Data will be made available on request.

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